ALAN TSENG

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EDUCATION

Master of Computer Science, National Taiwan University 2021-2024
Relevant Coursework: Blockchain and Big Data, Introduction to FinTech, Multimedia Security, Decentralized Applications Design and Practice, Computer Vision.
GPA: 4.23/4.3

EXPERIENCE

Software Engineer Intern 2022 - 2022
Line Corporation (Blockchain, React, Nx, Kubernetes, Swagger) Taipei, Taiwan

- Developed and deployed a Swagger-based API development tool, streamlining API development by 20% and ensuring a single source of truth. Seamless Kubernetes integration with optimized performance through DNS and load balancing configuration.
- Leveraged Nx to develop a scalable frontend architecture within a monorepo structure and collaboration with the Korean backend team.
- Led a team of 5 developers in an internal hackathon to build a merchant management system using NextJS and Spring Boot.
- Presented a talk about Ethereum and Decentralized Application Development within an internal study group.

PROJECTS

Decentralized Ticketing System
Master's Thesis (Solidity, TypeScript, React Native, Python, IPFS, GunDB, SQL, NextJS)

- Using lognormal distribution to simulate customer willingness to pay, developed three common ticketing market strategies and analyzed the different stakeholder gaming strategies in different sales strategies.
- Design and implementation of a decentralized ticketing system with P2P storage, achieving 90% code coverage via BDD testing and building a cross-platform app using React Native.

Human tremor symptoms detection
Academic Thesis (Python, Tensorflow)

- Developed an LSTM-RNN model with Tensorflow Functional API for tremor frequency and noise detection.
- Achieved an average accuracy of 94.8% on a dataset of 160,000 simulated signals.

Vaccine Tracking System
Course Project (Python, Solidity)

- Developed a blockchain-based vaccine tracking system utilizing Bloom Filter for efficient and rapid vaccine batch number verification with lower time complexity O(k) and space complexity O(m).

Drum.io, Game
Course Project (JavaScript, C, SocketIO, Arduino, Tensorflow.js)

- Designed and implemented a KNN-based multiplayer air drum game, awarded 1st place in Interactive Media Programming (52 participants) and 3rd place by collective vote (50 participants).
- Real-time analysis of acceleration data from an embedded device enabled intuitive air drumming and facilitated online competition.

Create your dancing robot, Computer Graphics
Course Project (C/C++, OpenGL)

- Got First place in the Computer graphics final project in a collective vote of 52 people.
- Implemented a model control system in OpenGL, enabling a robot to jump, inspired by Maya's graphical model control system.

Landing Planet, Game
Course Project (Java, Android)

- Won second place in a space exploration game development competition, with the support of 50 participants.
- Developed a space exploration game using Java and object-oriented concepts.

Rate My Professors TW, Full stack project
Side Project (NextJS, MVC, JWT, CI/CD, Web Crawling, NodeJS, MongoDB, PWA)

- Efficiently crawled course information from MCU-up and the school's course page using asyncio and aiohttp, reducing the crawling time by 83%.
- Built a course scheduler and course rating system using NextJS and implemented a REST API with FastAPI and MongoDB.

EXTRA-CURRICULAR ACTIVITIES

Developer Student Club Leader (2019-2021) - Advisor by Yuh-Pyng, Shieh, and Google Developers

- Selected as one of 800 leaders among over 3,000 passionate applicants worldwide for driving peer learning and community in a 2-year professional development program.
- Led and grew a technical student club, fostering a vibrant community and engaging over 73 attendees through talks on Deep Learning, PoseNet, Python & Colab, and Flutter.

ACM Competitive Programing Club Leader (2017-2021) - Supervised by Jeng-Sheng Yeh

- Enhanced programming skills through practice and problem-solving on platforms like UVa Online Judge.

Volunteered at exhibitions

- Provided technical support at the ART FUTURE exhibition showcasing blockchain applications.
- Assisted with Developer Student Club (DSC) and Google Developer Groups (GDG) events at SITCON(Student Information Technology Conference) and COSCUP(Conference for Open Source Coders, Users and Promoters).

HONORS

Research Excellence Award

- Recognized for outstanding research contributions, placing among the top 1% of graduating students.

Intel Andy Grove Scholarship

- Selected as one of 367 recipients worldwide, out of over 2,000 applicants, for demonstrating academic excellence and promising accomplishments.

College Student Research Scholarship

- Awarded a prestigious scholarship by the Ministry of Science and Technology, Taiwan.

PUBLICATIONS

- Sung, H.-M., Chen, T., **Tseng, H.-C.**, Prayogo, B., Lin, J.-Y., & Hung, Y.-P. (2023). **akaTick: Hybrid Mobile E-Ticketing System Based on Non-Fungible Tokens**. 2023 IEEE International Conference on Metaverse Computing, Networking and Applications (MetaCom), 686–687.

- **Tseng, H.-C**., Tu, C.-W., Huan, X.-Y., & Chia, T.-L. (2020). **Nighttime vehicle light detection based on deep learning and image enhancement**. 2020 IPPR Conference on Computer Vision, Graphics, and Image Processing (CVGIP), 1–8.

SKILLS

- Blockchain(EVM Compatible Chains(L1/L2), Tezos, COSMOS, Solana), Artificial Intelligence(Machine Learning, DeepLearning), Web(Frontend/Backend) Development, Cloud-Native Applications, Computer Vision, Socket, Chrome Extension, SSG/SSR, MVC,
- Python, C/C++, JavaScript/TypeScript, Solidity, Java, SmartPy, Swift, Dart, C#, Matlab
- FastAPI/Flask, NodeJS, React(NextJS, React Native), Tensorflow, Scikit-Learn, OpenCV, OpenGL, Numpy, Pandas, Matplotlib, Processing(p5), Flutter
- NoSQL(MongoDB), SQL(PostgreSQL, MySQL), p2p(orbitDB, GunJS), IPFS